Software Requirements Specification

for

Banquet Hall Booking System

Version 1.0

Prepared by

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Date: **13-03-2026**

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Draft Type and Number	Full Name	Information about the revision. This table does not need to be filled in whenever a document is touched, only when the version is being upgraded.	00/00/00

1 Introduction

This document presents the Software Requirements Specification (SRS) for the Banquet Hall Booking Management System developed for Hotel Sindhura East Court. The system aims to simplify the process of viewing hall availability, submitting booking requests, and managing bookings through an admin dashboard. It provides an online interface for customers to request bookings and allows administrators to approve, reject, and manage these requests efficiently. This section introduces the purpose of the document and outlines the key functionalities and structure of the system described in the following sections.

Document Purpose:
The purpose of this document is to clearly define the functional and non-functional requirements of the Banquet Hall Booking Management System. It serves as a reference for developers, project stakeholders, and evaluators to understand the system's features, design decisions, and expected behavior. This document also acts as a guideline for the implementation, testing, and future maintenance of the system

1.1 Document Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the **Banquet Hall Booking Management System**. The document provides a detailed description of the system's behavior, constraints, and operational environment to guide the design, development, testing, and deployment of the application.

This SRS serves as a reference for developers, project stakeholders, testers, and instructors to understand the system's requirements and expected functionality. The document outlines the system architecture, major components, user interactions, and technical requirements needed to implement the banquet hall booking platform. This specification applies to the initial release of the system and covers the complete functionality of the web-based booking platform.

1.2 Product Scope

The **Banquet Hall Booking Management System** is a web-based application designed to facilitate the booking and management of banquet halls in a hotel environment. The system enables users to view banquet hall details, check availability, and submit booking requests through an online interface. It also provides an administrative dashboard for hotel management to review booking requests, approve or reject reservations, and manage banquet hall availability.

The primary objective of this system is to streamline the banquet hall reservation process by replacing manual booking methods with a centralized digital platform. The system improves efficiency, reduces booking conflicts, and enhances the user experience by providing real-time availability information. Additionally, automated notifications ensure that users receive confirmation or rejection updates for their booking requests.

1.3 Intended Audience and Document Overview

This document is intended for several stakeholders involved in the development and evaluation of the Banquet Hall Booking Management System.

The primary audience includes **software developers**, who will use this document as a reference during system implementation, and **testers**, who will rely on the defined requirements to validate system functionality and performance. The **project manager and development team** will use the document to coordinate project tasks and ensure that all requirements are satisfied.

The document is also intended for **project evaluators and instructors**, who will review the system requirements to assess the completeness and feasibility of the proposed solution. Additionally, **hotel management stakeholders** may review the document to understand the system's capabilities and ensure that it aligns with operational needs.

The remainder of this SRS document is organized as follows:

- **Section 2 – Overall Description:** Provides a high-level overview of the system, including product perspective, user classes, operating environment, and design constraints.
- **Section 3 – System Features and Requirements:** Describes the functional requirements of the system and detailed system behaviors.
- **Section 4 – External Interface Requirements:** Defines user interfaces, software interfaces, and communication interfaces.
- **Section 5 – Non-Functional Requirements:** Describes system performance, security, reliability, and usability requirements.

Readers are encouraged to begin with the introduction and overall description sections before proceeding to the detailed requirement specifications.

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
API	Application Programming Interface
AWS	Amazon Web Services
CDN	Content Delivery Network
CRUD	Create, Read, Update, Delete
HTTPS	Hypertext Transfer Protocol Secure
JWT	JSON Web Token
REST	Representational State Transfer
RDS	Relational Database Service
S3	Simple Storage Service
SRS	Software Requirements Specification

UI	User Interface
UX	User Experience

1.5 Document Conventions

This document follows the **IEEE recommended structure for Software Requirements Specifications**. The formatting conventions used throughout this document are intended to maintain consistency and clarity.

The document uses **Arial font, size 12**, for all textual content, with section headings formatted according to the template provided. The text is **single-spaced with standard one-inch margins** on all sides.

Certain typographical conventions are used to improve readability and highlight important elements:

- **Bold text** is used for section headings and key terms.
- *Italic text* is used for explanatory comments and notes where applicable.
- Numbered lists are used to define functional requirements and system features clearly.
- Tables are used to present structured information such as definitions and abbreviations.

These conventions help maintain consistency throughout the document and ensure that the requirements are presented in a clear and structured manner.

1.6 References and Acknowledgments

This Software Requirements Specification (SRS) document was developed with reference to several academic resources, software engineering standards, and development tools that assisted in the analysis and design of the Banquet Hall Booking Management System.

The structure and formatting of this document follow the **IEEE 830 Software Requirements Specification standard**, which provides guidelines for organizing and documenting software requirements. Concepts related to software design, UML modeling, and requirement analysis were derived from course materials and standard software engineering practices discussed in class.

For system modeling and diagram creation, tools such as **Miro** and **Draw.io** were used to design UML diagrams including use case diagrams, class diagrams, and sequence diagrams. Development of the prototype web application utilized modern web technologies including **React.js, TypeScript, Node.js, and GitHub for version control**.

Additional references were consulted from online software development resources and documentation websites including:

- https://react.dev – Official React documentation
- https://developer.mozilla.org – Web development documentation
- https://nodejs.org – Node.js documentation
- https://github.com – Version control and collaboration platform

2 Overall Description

2.1 Product Overview

The **Banquet Hall Booking Management System** is a web-based application designed to digitize and streamline the process of reserving banquet halls for events. Traditionally in our hotel banquet hall reservations are handled manually through phone calls, physical booking registers, or direct communication with hotel management. These methods often lead to inefficiencies, booking conflicts, and lack of real-time availability information.

The proposed system replaces the manual booking process with a centralized online platform that allows customers to view available banquet halls, check availability, and submit booking requests through a web interface. The system also provides an administrative dashboard for hotel staff to review booking requests, approve or reject reservations, and manage banquet hall schedules.

The application is designed as a **self-contained web-based system** consisting of a frontend interface, backend server, and a relational database. Users interact with the system through a web browser, which communicates with the backend server through secure REST APIs. The backend server processes requests, manages authentication, enforces booking rules, and stores data in the system database. The system also integrates notification services to inform users about booking confirmations or rejections.

At a high level, the system operates within the context of a hotel management environment where users request bookings and administrators manage the scheduling and availability of banquet halls.

2.2 Product Functionality

The Banquet Hall Booking Management System provides several core functionalities that support both customer booking operations and administrative management of banquet halls.

The major functions of the system include:

- Allow users to **register and log in** to the system securely.
- Allow users to **view banquet hall details**, including capacity, features, and pricing.
- Enable users to **check banquet hall availability** based on selected dates.
- Allow users to **submit booking requests** for available halls.
- Prevent **double booking conflicts** through availability validation.
- Allow administrators to **review booking requests** submitted by users.
- Allow administrators to **approve or reject booking requests**.
- Enable administrators to **manage banquet hall availability schedules**.

- Store and manage all system data including **users, halls, and bookings** in a relational database.
- Send **notifications to users** when booking requests are approved or rejected.

These functions ensure that the booking process becomes more efficient, transparent, and manageable for both customers and hotel management.

2.3 Design and Implementation Constraints

The development of the Banquet Hall Booking Management System is subject to several design and implementation constraints that influence the architecture, tools, and technologies used.

First, the system will be implemented as a **web-based application** using modern web development technologies. The frontend interface will be developed using **React.js with Vite as the build tool and Tailwind CSS for styling**, while the backend will be developed using **Node.js and the Express.js framework**. The system database will be implemented using **MySQL**, which provides structured relational data storage.

Second, the system must follow **secure authentication practices**, including the use of **JSON Web Tokens (JWT)** for user authentication and **bcrypt hashing** for password encryption. All communication between the frontend and backend will occur through **HTTPS-based REST APIs**.

Additionally, the project uses **UML (Unified Modeling Language)** for modeling system components, interactions, and behaviors. UML diagrams such as **use case diagrams, class diagrams, sequence diagrams, and system architecture diagrams** will be used to represent the system design.

References:

- Gomaa, H. *Designing Concurrent, Distributed, and Real-Time Applications with UML.* Addison-Wesley.
- IEEE. *UML Specification.*
- Gomaa, H. *Software Modeling and Design: UML, Use Cases, Patterns, and Software Architectures.*

Finally, the system must operate within standard web browser environments and support common browsers such as **Google Chrome, Microsoft Edge, and Mozilla Firefox**.

2.4 Assumptions and Dependencies

Several assumptions and dependencies have been made during the design of the Banquet Hall Booking Management System. These assumptions influence the development approach and system functionality.

The following assumptions are made:

- The hotel management provides accurate and updated information regarding **banquet hall details, pricing, and availability schedules**.
- Users accessing the system have **internet connectivity and a modern web browser**.
- The hosting environment will support **Node.js runtime and MySQL database services**.
- Users will provide valid personal information when registering for booking services.
- The number of simultaneous users will remain within the capacity supported by the hosting infrastructure.

The system also depends on several external components and services, including:

- **Email or SMS notification services** for sending booking confirmation or rejection alerts.
- **Cloud hosting services** such as AWS for deploying the application infrastructure.
- **Third-party libraries and frameworks**, including React.js, Express.js, and authentication libraries.

If these assumptions change or external dependencies become unavailable, the system design or functionality may require modification.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The Banquet Hall Booking Management System provides a **web-based user interface** that allows both customers and administrators to interact with the system through a web browser. The interface is designed to be simple, intuitive, and accessible on both desktop and mobile devices.

The main user interfaces of the system include:

1. Home Page Interface

- Displays available banquet halls.
- Shows hall images, descriptions, and capacity information.
- Includes a **“Book Now” button** to initiate the booking process.

2. Login Interface

- Allows users and administrators to log into the system.
- Accepts user credentials such as email and password.
- Redirects administrators to the **Admin Dashboard** upon successful authentication.

3. Admin Dashboard Interface

- Allows administrators to:
  - Review booking requests.
  - Approve or reject booking requests.
  - Add hall availability.
  - Approve or reject new admin registration requests.

4. Admin Registration Interface

- Allows new administrators to request admin access.
- Existing administrators must approve the request before the new admin gains access.

Users interact with the system through **web forms, buttons, and menu navigation** using a mouse, keyboard, or touch interface.

3.1.2 Hardware Interfaces

The system does not require specialized hardware and operates on standard computing devices.

Supported hardware interfaces include:

- Desktop computers
- Laptop computers
- Tablets
- Smartphones
- Internet-enabled devices with a web browser

Users interact with the system through input devices such as:

- Keyboard
- Mouse
- Touchscreen

The server hosting the application communicates with the database server through standard network connections.

3.1.3 Software Interfaces

The Banquet Hall Booking Management System interacts with the following software components:

Frontend Interface

- Built using **React.js**
- Styled using **CSS or Tailwind CSS**
- Communicates with backend services using REST APIs

Backend Interface

- Developed using **Node.js and Express.js**
- Handles business logic, authentication, and booking management

Database Interface

- Uses **MySQL** relational database
- Stores data related to:
  - Users
  - Admin accounts
  - Banquet halls
  - Booking requests
  - Availability schedules

Testing Interface

- API testing performed using **Postman**

Version Control

- Source code managed using **GitHub**

3.2 Functional Requirements

The following functional requirements define the expected behavior of the system.

3.2.1 User Registration

The system shall allow new administrators to submit registration requests.

3.2.2 User Authentication

The system shall allow users and administrators to log into the system using valid credentials.

3.2.3 View Banquet Halls

The system shall display a list of available banquet halls along with images, descriptions, and capacity information.

3.2.4 Submit Booking Request

The system shall allow users to submit booking requests for a selected banquet hall.

3.2.5 View Booking Requests

The system shall allow administrators to view all pending booking requests.

3.2.6 Approve Booking Requests

The system shall allow administrators to approve booking requests.

3.2.7 Reject Booking Requests

The system shall allow administrators to reject booking requests.

3.2.8 Manage Hall Availability

The system shall allow administrators to add and update banquet hall availability.

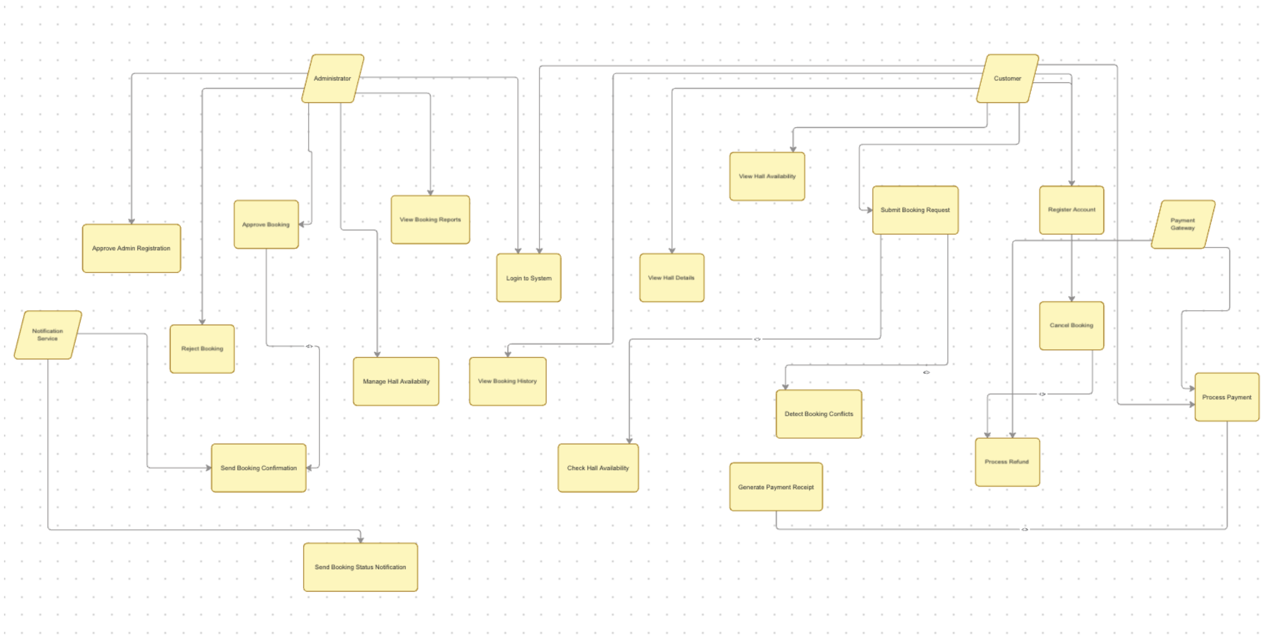
3.2.9 Admin Registration Approval

The system shall allow administrators to approve or reject new admin registration requests.

3.2.10 Update Booking Status

The system shall update the status of a booking request after admin approval or rejection.

3.3 Use Case Model



3.3.1 Use Case U1 – User Registration

Author: Pranav Reddy

Purpose
Allows a new user to create an account in the Banquet Hall Booking System so they can access booking services.

Requirements Traceability
F1 – User Registration
F2 – User Authentication

Priority
High

Preconditions

- User must have access to the system website.
- User must provide valid personal information.

Postconditions

- A new user account is successfully created in the system database.

Actors

- User
- System

Extends

None

Flow of Events

Basic Flow

1. User opens the registration page.
2. User enters registration details (name, email, password).
3. System validates the input data.
4. System checks if the email already exists in the database.
5. If the email does not exist, the system saves the user details.
6. System confirms successful registration.

Alternative Flow

- If the email already exists, the system displays an error message.

Exceptions

- Database connection failure.

Includes

None

Notes/Issues

Password encryption must be implemented for security.

3.3.2 Use Case U2 – User Login

Author: Sathwik Reddy

Purpose

Allows registered users to log into the system to access booking features.

Requirements Traceability

F2 – User Authentication

Priority

High

Preconditions

- User must have an existing account.

Postconditions

- User gains access to the system dashboard.

Actors

- User
- System

Extends

None

Flow of Events

Basic Flow

1. User enters email and password.
2. System sends credentials to the database.
3. Database validates the credentials.
4. System grants access to the user.

Alternative Flow

- If credentials are incorrect, the system displays an error message.

Exceptions

- Authentication server failure.

Includes

None

Notes/Issues

Multiple failed login attempts may trigger security measures.

3.3.3 **Use Case U3 – View Hall Availability**

Author: Jyotsna Pandranki

Purpose

Allows users to view available banquet halls and their booking availability.

Requirements Traceability

F3 – View Hall Information
F4 – Check Hall Availability

Priority

High

Preconditions

- User must be logged into the system.

Postconditions

- Available halls are displayed to the user.

Actors

- User
- System

Extends

None

Flow of Events

Basic Flow

1. User requests the list of available halls.
2. System sends request to the hall database.
3. Hall database returns hall information.
4. System checks availability for selected date/time.
5. System displays available halls.

Alternative Flow

- If no halls are available, the system displays a notification.

Exceptions

- Database retrieval failure.

Includes

Availability Check (U4)

Notes/Issues

Hall capacity and amenities should be displayed for user decision-making.

3.3.4 **Use Case U4 – Submit Hall Booking Request**

Author: Jyotsna Pandranki

Purpose

Allows users to request booking for a specific hall on a selected date and time.

Requirements Traceability

F5 – Submit Booking Request
F6 – Store Booking Information

Priority

High

Preconditions

- User must be logged in.
- Hall must be available.

Postconditions

- Booking request is stored in the system database.

Actors

- Customer
- Booking System
- Admin

Extends

None

Flow of Events

Basic Flow

1. Customer selects hall, date, and time.
2. System checks hall availability.
3. System confirms availability.
4. Customer submits booking request.
5. System saves booking details in the database.
6. System notifies the administrator.

Alternative Flow

- If hall is unavailable, system suggests alternate slots.

Exceptions

- Booking database failure.

Includes

View Hall Availability (U3)

Notes/Issues

Booking requests remain in **Pending status** until admin approval.

3.3.5 **Use Case U5 – Admin Booking Approval**

Author: Akshara Parikibanda

Purpose

Allows the administrator to approve or reject booking requests submitted by users.

Requirements Traceability

F7 – Approve Booking
F8 – Reject Booking

Priority

High

Preconditions

- Admin must be logged in.
- Pending booking requests must exist.

Postconditions

- Booking status updated in database.

Actors

- Admin
- System
- Notification Service

Extends

None

Flow of Events

Basic Flow

1. Admin logs into the system.
2. Admin views pending booking requests.
3. System retrieves requests from database.
4. Admin selects approve or reject option.
5. System updates booking status.
6. System notifies customer of decision.

Alternative Flow

- Admin may postpone decision.

Exceptions

- Database update failure.

Includes

None

Notes/Issues

Approved bookings block the selected hall time slot.

3.3.6 **Use Case U6 – Payment Processing**

Author: Yeshwanth Ram Gudavalli

Purpose

Allows customers to complete payment for hall bookings.

Requirements Traceability

F9 – Payment Processing
F10 – Payment Recording

Priority

Medium

Preconditions

- Booking must be approved.
- Customer must have a valid payment method.

Postconditions

- Payment transaction stored in database.

Actors

- Customer
- Payment Gateway
- System

Extends

None

Flow of Events

Basic Flow

1. Customer initiates payment.
2. System sends payment request to payment gateway.
3. Gateway processes transaction.
4. Payment result returned to system.
5. System stores payment record.
6. System generates invoice.

Alternative Flow

- Payment failure due to insufficient funds.

Exceptions

- Payment gateway service unavailable.

Includes

Booking Request (U4)

Notes/Issues

Payment receipts should be downloadable.

3.3.7 Use Case U7 – Booking Cancellation

Author: Yuthika Reddy

Purpose

Allows customers to cancel an existing booking and receive a refund if applicable.

Requirements Traceability

F11 – Cancel Booking
F12 – Refund Payment

Priority

Medium

Preconditions

- Booking must exist.
- Cancellation must occur before event date.

Postconditions

- Booking status updated to cancelled.
- Refund processed if applicable.

Actors

- Customer
- System
- Payment System

Extends

None

Flow of Events

Basic Flow

1. Customer requests booking cancellation.
2. System verifies booking information.
3. System updates booking status to cancelled.
4. System initiates refund process.
5. Payment system processes refund.
6. System confirms cancellation to customer.

Alternative Flow

- Refund not applicable if cancellation occurs after deadline.

Exceptions

- Payment system failure.

Includes

Payment Processing (U6)

Notes/Issues

Refund policies must be defined by the hotel management.

4 Other Non-functional Requirements

4.1 Performance Requirements

The Banquet Hall Booking Management System must provide efficient performance to ensure a smooth user experience for both customers and administrators.

P1. Page Load Performance
The system shall load the homepage and banquet hall listings within **3 seconds** under normal network conditions.

P2. Concurrent Users
The system shall support at least **50 simultaneous users** accessing the system without noticeable performance degradation.

P3. Booking Processing Time
The system shall process booking requests and update the booking database within **2 seconds** after submission.

P4. Admin Dashboard Response Time
The system shall display booking requests and admin registration requests on the admin dashboard within **2 seconds** after the administrator logs in.

P5. Database Query Performance
The system shall retrieve booking and hall availability data from the database within **1 second** under normal operating conditions.

These performance requirements ensure that users experience minimal delays while interacting with the system.

4.2 Safety and Security Requirements

The Banquet Hall Booking Management System must ensure that user data, booking information, and payment transactions are protected from unauthorized access or misuse.

S1. User Authentication
The system shall require users and administrators to authenticate using valid login credentials before accessing restricted system features.

S2. Password Protection
User passwords shall be stored using secure hashing techniques (e.g., bcrypt encryption) to prevent unauthorized access.

S3. Secure Communication
All communication between the client application and the server shall use **HTTPS encryption** to protect data transmitted over the internet.

S4. Role-Based Access Control
The system shall implement role-based access control to ensure that only administrators can approve bookings, manage hall availability, and approve admin registrations.

S5. Data Privacy Protection
The system shall protect sensitive user data such as email addresses, booking information, and payment records from unauthorized access.

S6. Session Management
User sessions shall automatically expire after a period of inactivity to prevent unauthorized system access.

S7. Payment Security
Payment transactions shall be processed through a secure payment gateway that complies with standard online payment security protocols.

4.3 Software Quality Attributes

Software quality attributes describe the characteristics that ensure the system is reliable, maintainable, and user-friendly.

4.3.1 Reliability

The system shall operate reliably under normal operating conditions and maintain consistent service availability.

R1. System Availability
The system shall maintain at least **99% uptime** excluding scheduled maintenance.

R2. Error Handling
The system shall provide clear error messages to users in the event of invalid inputs or system failures.

R3. Data Integrity
The system shall ensure that booking data and payment information are stored accurately and protected from corruption.

4.3.2 Usability

The system shall provide a simple and intuitive user interface that allows users to complete tasks efficiently.

U1. User-Friendly Interface
The system shall provide clear navigation, buttons, and instructions to allow users to perform booking operations easily.

U2. Minimal Learning Curve
Users should be able to understand and use the booking interface with minimal training.

U3. Responsive Design
The system shall support responsive design so that the website can be accessed on desktops, tablets, and mobile devices.

4.3.3 Maintainability

The system shall be designed in a way that allows future updates and improvements with minimal effort.

M1. Modular Architecture
The system shall be developed using a modular architecture separating frontend, backend, and database components.

M2. Code Documentation
Developers shall maintain proper documentation and comments in the code to assist future maintenance.

M3. Version Control
The project source code shall be managed using **GitHub** to enable version tracking and collaborative development.

4.4 Other Requirements

4.4.1 Database Requirements

The system shall use a **relational database management system (RDBMS)** to store system data. The database will contain tables for:

- Users
- Admin accounts
- Banquet halls
- Booking records
- Payment transactions
- Hall availability schedules

The database shall enforce **primary keys, foreign key constraints, and data validation rules** to maintain data integrity.

4.4.2 Deployment Requirements

The system shall be deployable on cloud infrastructure such as **Amazon Web Services (AWS)** or similar hosting platforms.

Possible deployment components include:

- **Web Server** for hosting the application
- **Database Server** for storing system data
- **Cloud storage services** for static resources such as images

4.4.3 Development Tools

The system shall be developed using the following technologies:

Frontend:

- React.js
- HTML, CSS, JavaScript

Backend:

- Node.js
- Express.js

Database:

- MySQL or PostgreSQL

Development Tools:

- Visual Studio Code
- GitHub (version control)
- Postman (API testing)

5 Other Requirements

This section describes additional requirements that support the long-term operation, maintainability, and usability of the Banquet Hall Booking Management System.

5.1 System Maintenance and Update Management

The system shall support regular maintenance and updates without causing major disruptions to the users.

OR1. Version Management
The system source code shall be managed using a version control system such as **GitHub**, allowing developers to track changes, revert to previous versions, and manage collaborative development.

OR2. Deployment Process
System updates shall be deployable using a structured deployment process to ensure that updates do not affect existing system functionality.

OR3. System Monitoring
Administrators shall be able to monitor the system status and performance metrics to identify potential issues during operation.

OR4. Backup and Recovery
The system shall automatically create periodic backups of the database to prevent data loss in case of system failure.

5.2 Administrative Booking Interface

The system shall provide an administrative interface that allows hotel staff to manage bookings directly from the hotel premises.

OR5. Walk-in Booking Management
Administrators shall be able to create bookings on behalf of walk-in customers directly through the admin dashboard.

OR6. Real-Time Availability View
The system shall allow administrators to view the availability of banquet halls in real time before creating bookings.

OR7. Booking Modification
Administrators shall be able to modify or update existing bookings when requested by customers.

OR8. Booking Cancellation by Admin
Administrators shall have the authority to cancel bookings when necessary.

5.3 Booking History and Reporting

The system shall provide tools for administrators to track and analyze booking data.

OR9. Booking History Access
Administrators shall be able to view a list of both **current bookings and past bookings**.

OR10. Booking Search and Filtering
The system shall allow administrators to search and filter bookings by date, customer name, hall name, or booking status.

OR11. Booking Reports
The system shall allow administrators to generate reports related to bookings, hall utilization, and revenue.

5.4 Notification and Communication Features

The system shall provide automated notifications to inform users about booking updates.

OR12. Booking Confirmation Notifications
Customers shall receive notifications when their booking requests are approved or rejected.

OR13. Payment Confirmation Notifications
The system shall notify customers after successful payment transactions.

OR14. Cancellation Notifications
Customers shall receive notifications if their booking has been cancelled.

5.5 Future Scalability and System Expansion

The system shall be designed to support future expansion and integration with additional services.

OR16. Mobile Application Integration
The system should allow future integration with a mobile application for easier customer access.

OR17. Third-Party Integration
The system should support future integration with third-party services such as online payment gateways, email services, and SMS notification systems.

Appendix A – Data Dictionary

The data dictionary defines the key data elements, variables, and system states used within the Banquet Hall Booking Management System. It provides descriptions of each data element, their possible values, and their role within the system operations.

Data Element	Type	Description	Possible Values / Format	Used In / Related Operations
userId	Integer	Unique identifier for each registered user	Auto-generated numeric ID	User registration, login, booking creation

fullName	String	Name of the registered user	Alphabetic characters	User profile management
email	String	Email address used for login and communication	Valid email format	Registration, login, notifications
passwordHash	String	Encrypted password stored in the database	Hashed string value	User authentication
role	Enum	Defines user access level	Customer, Admin	Access control, system permissions
hallId	Integer	Unique identifier for each banquet hall	Numeric ID	Hall information management
hallName	String	Name of the banquet hall	Text value	Hall listing and booking selection
capacity	Integer	Maximum number of guests allowed in the hall	Numeric value	Booking validation
location	String	Physical location of the banquet hall	Text description	Hall information display
amenities	String[]	List of facilities provided in the hall	Array of text values	Hall description
bookingId	Integer	Unique identifier for each booking	Auto-generated numeric ID	Booking management
bookingDate	Date	Date on which the hall is booked	Date format	Booking scheduling
eventType	String	Type of event for the booking	Wedding, Conference, Party, etc.	Booking details
guestCount	Integer	Number of guests expected	Numeric value	Capacity validation
bookingStatus	Enum	Status of the booking request	Pending, Approved, Rejected, Cancelled	Booking lifecycle
paymentId	Integer	Unique identifier for payment transaction	Numeric ID	Payment tracking
paymentAmount	Double	Total amount paid for booking	Currency value	Payment processing
paymentMethod	String	Method used for payment	Credit Card, Debit Card, UPI, Cash	Payment processing
paymentStatus	Enum	Status of payment transaction	Pending, Completed, Failed, Refunded	Payment management
availabilityId	Integer	Unique identifier for hall availability entry	Numeric ID	Availability tracking
availabilityDate	Date	Date for which hall availability is checked	Date format	Availability management
isAvailable	Boolean	Indicates if a hall is available for booking	True / False	Availability check
adminRequestId	Integer	Unique identifier for admin registration requests	Numeric ID	Admin approval workflow
adminRequestStatus	Enum	Status of admin registration request	Pending, Approved, Rejected	Admin management

notificationMessage	String	Message sent to users regarding booking updates	Text value	Notification service
transactionReference	String	Unique reference for payment transaction	Alphanumeric string	Payment verification

System State Variables

State Variable	Description	Possible States
Booking Status	Indicates current stage of a booking request	Pending, Approved, Rejected, Cancelled
Payment Status	Indicates current stage of payment processing	Pending, Completed, Failed, Refunded
Hall Availability	Indicates whether a hall can be booked for a specific time slot	Available, Not Available
User Authentication Status	Indicates if a user is logged in	Logged In, Logged Out

System Outputs

Output	Description
Registration Confirmation	Notification confirming successful user registration
Login Result	Authentication success or failure message
Hall Availability List	Display of available halls and their details
Booking Confirmation	Notification confirming booking submission
Booking Status Notification	Message informing user of booking approval or rejection
Payment Receipt	Confirmation of successful payment transaction
Cancellation Confirmation	Notification confirming booking cancellation

Appendix B - Group Log

During the development of the Banquet Hall Booking Management System project we conducted multiple group discussions and collaborative work sessions to plan, design, and document the system requirements. In the initial meetings we focused understanding the overall objectives of the system. We worked together to identify the major stakeholders, define the system scope, and gather both functional and non-functional requirements. We also discussed how the system would operate from both the customer and administrator perspectives.

As the project progressed several meetings were held to design the technical aspects of the system. Then we identified key subsystems such as the User Management Subsystem, Hall Information Subsystem, Booking Management Subsystem, Payment Processing Subsystem, Admin Control Subsystem, and Availability & Conflict Detection Subsystem. Based on these subsystems we collaboratively developed the system architecture and UML diagrams including the class diagram, sequence diagrams, and use case diagrams. Each one contributed to different parts of the documentation and the results were reviewed together.

The team also refined the software requirements specification by discussing system behavior, performance expectations, security considerations, and administrative functionality. Special attention was given to designing features that would help hotel staff manage bookings efficiently, including the ability for administrators to approve booking requests, manage hall availability, and handle walk-in customers. We reviewed the entire document together corrected any mistakes and ensured that all sections of the SRS were properly completed before submission.

Throughout the project the group maintained continuous communication and coordination to ensure steady progress and equal contribution from all members. This collaborative approach helped the team successfully complete the system analysis and documentation for the banquet hall booking system